Class 10 · Science

Chapter 2: Acids, Bases and Salts

Worksheet — Acids, Bases and Salts (Class 10 CBSE Science)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions. Show working / write balanced equations. Total time: 50 minutes.

Q1. Litmus paper turns ____ in an acid. [1 mark]

- (A) red
- (B) blue
- (C) yellow
- (D) green

Q2. All acids release OH■ ions in water. [1 mark]

- (A) True
- (B) False

Q3. Pure water has a pH of ____. [1 mark]

Q4. Which household acid is found in vinegar? [1 mark]

- (A) citric acid
- (B) acetic acid
- (C) lactic acid
- (D) sulphuric acid

Q5. When dilute HCl is added to limestone (CaCO■), the gas produced turns lime water: [2 marks]

- (A) yellow
- (B) milky / white
- (C) blue
- (D) green

Q6. Write balanced equations for the reactions of dilute HCl with: (a) zinc metal, (b) calcium carbonate, (c) copper(II) oxide. [3 marks]

Q7. Distinguish between (i) STRONG and WEAK acids, (ii) CONCENTRATED and DILUTE solutions. Give one example of each combination. [3 marks]

Q8. Why must acid always be added to water and never the other way around? What can happen if done wrong? [3 marks]

Q9. Why does tooth decay start at low pH? What kind of substances should we use to clean our teeth? [3 marks]

Q10. A student adds dilute hydrochloric acid to four substances: (a) sodium carbonate, (b) zinc granules, (c) copper turnings, (d) copper(II) oxide. For each, predict whether a reaction occurs and write the balanced equation. State an observation that indicates the reaction has happened. [6 marks]

Q11. Common salt (NaCl) is the raw material for several industrial chemicals. Name FOUR products that can be made from common salt and the reactions involved. Mention one use of each product. [6 marks]

Answer Key

Worksheet — Acids, Bases and Salts (Class 10 CBSE Science)

Q1. Litmus paper turns ____ in an acid.

Answer: red

Standard rule.

MCQ · EASY · 10-CBSE-SCI-ABS-01

Q2. All acids release OH■ ions in water.

Answer: False

Acids release H■; bases release OH■.

TRUE_FALSE · EASY · 10-CBSE-SCI-ABS-04

Q3. Pure water has a pH of ____.

Answer: 7

Neutral.

FILL_BLANK · EASY · 10-CBSE-SCI-ABS-06

Q4. Which household acid is found in vinegar?

Answer: acetic acid

Vinegar = ~5% acetic acid in water.

MCQ · EASY · 10-CBSE-SCI-ABS-01

Q5. When dilute HCl is added to limestone (CaCO■), the gas produced turns lime water:

Answer: milky / white

CO■ + Ca(OH)■ → CaCO■↓ + H■O.

MCQ · MEDIUM · 10-CBSE-SCI-ABS-02

Q6. Write balanced equations for the reactions of dilute HCl with: (a) zinc metal, (b) calcium carbonate, (c) copper(II) oxide.

Answer: (a) Zn + 2 HCl → ZnCl■ + H■↑. (b) 2 HCl + CaCO■ → CaCl■ + H■O + CO■↑. (c) 2 HCl + CuO → CuCl■ + H■O.

Three classic acid reactions.

SHORT_ANSWER · MEDIUM · 10-CBSE-SCI-ABS-02

Q7. Distinguish between (i) STRONG and WEAK acids, (ii) CONCENTRATED and DILUTE solutions. Give one example of each combination.

Answer: (i) STRONG acid ionises COMPLETELY in water (e.g. HCl, H■SO■); WEAK acid ionises PARTIALLY (e.g. acetic, citric). (ii) CONCENTRATED has a high amount of solute per litre; DILUTE has a low amount. The two ideas are independent — examples: a dilute strong acid (HCl 0.01 M) is dilute and strong; concentrated acetic acid (glacial) is concentrated and weak.

Strength vs concentration.

SHORT_ANSWER · MEDIUM · 10-CBSE-SCI-ABS-04

Q8. Why must acid always be added to water and never the other way around? What can happen if done wrong?

Answer: Diluting concentrated acid is HIGHLY EXOTHERMIC — heat is released as ions hydrate. If water is added to a concentrated acid, the small amount of water heats up rapidly to boiling and may cause the acid to SPLASH OUT and cause burns. Adding acid to a much larger volume of water lets heat be dispersed safely.

Standard safety rule.

SHORT_ANSWER · MEDIUM · 10-CBSE-SCI-ABS-05

Q9. Why does tooth decay start at low pH? What kind of substances should we use to clean our teeth?

Answer: Bacteria in the mouth ferment sugars to acid. When the mouth's pH falls BELOW 5.5, the calcium phosphate of tooth enamel begins to dissolve — this is the start of tooth decay. To clean teeth and to neutralise the excess acid, we use BASIC TOOTHPASTES (which contain mild bases like sodium hydrogencarbonate and calcium carbonate). They restore the pH above 5.5 and stop the decay.

Practical pH application.

SHORT_ANSWER · MEDIUM · 10-CBSE-SCI-ABS-07

Q10. A student adds dilute hydrochloric acid to four substances: (a) sodium carbonate, (b) zinc granules, (c) copper turnings, (d) copper(II) oxide. For each, predict whether a reaction occurs and write the balanced equation. State an observation that indicates the reaction has happened.

Answer: (a) Sodium carbonate Na■CO■: reacts. 2 HCl + Na■CO■ → 2 NaCl + H■O + CO■↑. Observation: rapid effervescence; gas turns lime water milky. (b) Zinc Zn: reacts. Zn + 2 HCl → ZnCl■ + H■↑. Observation: bubbles of colourless gas (H■); burns with a 'pop'. (c) Copper Cu: NO reaction with dilute HCl — copper is below hydrogen in the reactivity series. (d) Copper(II) oxide CuO: reacts. 2 HCl + CuO → CuCl■ + H■O. Observation: black CuO dissolves to form a blue-green solution.

Predict + balance + observe.

LONG_ANSWER · HARD · 10-CBSE-SCI-ABS-02

Q11. Common salt (NaCl) is the raw material for several industrial chemicals. Name FOUR products that can be made from common salt and the reactions involved. Mention one use of each product.

Answer: (1) Sodium hydroxide (NaOH) — by electrolysis of brine: 2 NaCl + 2 H■O → 2 NaOH + Cl■ + H■. Use: soap making. (2) Chlorine (Cl■) — same chlor-alkali process. Use: PVC, water disinfection. (3) Bleaching powder (CaOCl■): Ca(OH)■ + Cl■ → CaOCl■ + H■O. Use: bleaching cotton / paper, disinfection of drinking water. (4) Baking soda (NaHCO■): NaCl + H■O + NH■ + CO■ → NaHCO■ + NH■Cl. Use: baking, antacid, fire extinguishers. (Bonus: washing soda Na■CO■·10 H■O — used in detergents and glass.)

Industrial chemistry from common salt.

LONG_ANSWER · HARD · 10-CBSE-SCI-ABS-08